

China Renewable Energy — Industry Fundamental Analysis

How the industry works, where growth comes from, and what structurally limits it

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Method & source discipline. The official State Council Information Office briefing on the draft 15th Five-Year Plan is now used as a primary source for plan-related claims, cited by page. All other claims remain tied to named external sources — reputable industry reports, regulator and industry-body publications, and major-player results — listed at the end. Note the briefing interprets the *draft* Outline, so it confirms some points directly and is silent on others; the latter keep their external citations. Sources are paraphrased rather than quoted at length. The most recent data available is used; older data is flagged in text. Inferences are marked explicitly and are not presented as fact.

1. Industry Purpose & Core Economics

China's renewable industry exists to do two things at once: supply a fast-growing, electrifying economy with power, and reduce dependence on imported fossil fuels. By end-2025 it had become the backbone of the system — wind and solar reached about 1,840 GW of grid-connected capacity, roughly 47.3% of total installed power capacity, surpassing thermal power for the first time, and accounted for about 22% of electricity output (NEA, Feb 2026). On a broader basket, the NDRC briefing states renewable energy (including hydro) reached about 60% of total capacity over the 14th-Plan period [15th FYP transcript, p.6] — the two figures are consistent, measuring different baskets (wind + solar vs all renewables).

The economics split into two layers that behave very differently. The first is **manufacturing** — polysilicon, wafers, cells, modules, turbines, batteries — where value is created through scale and cost leadership. The second is **power generation**, owning and operating plants, where value is captured through long-term electricity revenue. In 2025 the split was stark: manufacturing was largely loss-making while generation stayed profitable. LONGi, a leading module maker, reported that its power-plant operations were its only profitable segment (about 14.4% gross margin) while module products ran a negative margin (pv magazine, Sep 2025).

2. Industry Structure & Competitive Edge

The structure is a high-volume manufacturing base feeding both domestic and global demand — concentrated at the top, crowded below. The top four module makers (JinkoSolar, LONGi, Trina, JA Solar) accounted for roughly 58% of about 536 GW of global module shipments in 2025 (InfoLink via TaiyangNews, Mar 2026). In polysilicon — the purified silicon feedstock — concentration is higher, with the top five producers holding over 80% of output (Pyramids & Pagodas, Nov 2025).

The competitive edge is cost and innovation built on scale, supported by centralized planning, a domestic supply chain, and state-backed financing (CarbonCredits, Feb 2026). On technology, China's share of global solar patent applications reached around 65% by 2024 (CPIA report, via CSIS, Mar 2026). China also ranks first in the world in offshore wind power [15th FYP transcript, p.21]. The durable advantage is the integrated supply chain plus relentless cost reduction — not brand or proprietary lock-in.

3. Demand & Growth Drivers

Growth comes from three sources. **Domestic electricity demand** keeps rising with the electrification of transport, heating and industry; demand is expected to exceed 13 trillion kWh by 2030, growing 4–6% a year from 2025 (kr-asia, May 2025).

Policy targets provide the anchor: President Xi's 2020 pledge of 1,200 GW of wind and solar by 2030 was hit years early (inferred from the 2025 capacity figures in Section 1 against that target; Bloomberg / Reuters,

2020–2023). **Exports** remain large — Chinese module exports were about 267.6 GW in 2025, up 13% year-on-year (InfoLink via TaiyangNews, Mar 2026).

4. Supply Side, Cost Structure & Constraints

The defining feature of 2024–2025 was **overcapacity** — far more manufacturing capacity than demand. Polysilicon prices collapsed from about \$32.7/kg in February 2023 to \$4.4/kg by May 2024 (pv-tech, Dec 2025), utilization fell to 50–60%, and prices were down more than 80% from 2023 (Pyramids & Pagodas, Nov 2025). The result was sector-wide losses: the top four module makers posted combined net losses of nearly RMB 11bn (\$1.54bn) in H1 2025, about two-and-a-half times the prior year (pv-tech, Sep 2025).

Two physical constraints sit on top of the financial one. **Curtailment** — power generated but not used because the grid cannot absorb it — is rising in the resource-rich northwest, with reported rates in Gansu above 30% for solar and higher for wind (Dialogue Earth, Mar 2026). One mitigation now in the plan is to move eligible energy-intensive industries to regions with high concentrations of renewable resources, and to lift efficiency in computing-power facilities and 5G base stations [15th FYP transcript, p.7] — co-locating demand with desert and Gobi supply. The other constraint is grid flexibility, which China is investing heavily to fix: more than \$80bn went into grid infrastructure in 2024, including ultra-high-voltage lines linking remote generation to coastal demand (WEF, Dec 2025) [data is from 2024].

5. Technology Changes

The cell-technology base has shifted quickly. As of early 2026, TOPCon accounted for almost 85% of global production while older PERC fell below 5% (CRU via pv-tech, Mar 2026). The next frontier is tandem cells — stacking a perovskite layer on silicon to break the single-material efficiency ceiling. Recent milestones include Trina's full-size TOPCon-perovskite tandem module at 32.6% efficiency (Feb 2026) and LONGi's 33.35% flexible perovskite-silicon tandem cell, certified by the US NREL (Nov 2025).

Commercial perovskite remains early-stage: no major commercial perovskite modules were on sale in 2026, with launches targeted for 2026–2028 (SurgePV, Mar 2026).

6. Regulatory Changes — the 15th Five-Year Plan (2026–2030)

The 15th Five-Year Plan, presented at the March 2026 'Two Sessions', sets the regulatory direction. The official briefing frames decarbonisation around carbon peaking and neutrality and confirms a shift from controlling energy use to a dual-control system covering both total carbon emissions and emission intensity [15th FYP transcript, p.6]. Separately, analysts report a headline 17% cut in carbon intensity over 2026–30 — a number not given in the briefing itself — which they note is modestly weaker than the 14th plan's 18% (Sightline / CREA, Mar 2026).

On energy specifically, the plan calls for simultaneous development of wind, solar, hydro and nuclear, a decade-long campaign to double the share of non-fossil energy, and construction of a 'new type of power system' [15th FYP transcript, p.6, p.17]. A precision point that matters for investors: 'non-fossil' explicitly includes nuclear, so it is broader than 'renewables' and the two should not be read as interchangeable. Non-fossil energy is targeted at 25% of total consumption by 2030 [15th FYP transcript, p.17]; a further 30%-by-2035 figure comes from external analysis, not this briefing (CREA, Mar 2026).

The most important market change predates the plan. **Document 136** (Feb 2025) ended guaranteed fixed prices: all renewable projects commissioned after 1 June 2025 trade in the power market, storage-pairing mandates are removed, and contracts-for-difference replace fixed tariffs (pv magazine, Jun 2025). Implementation is uneven — as of mid-October 2025 only 18 provinces had finalised their Document 136 plans (Carbon Brief, Oct 2025).

Deployment-focused tools include about 100 national zero-carbon parks and more than 10,000 km of zero-carbon transportation corridors [15th FYP transcript, p.7]. Externally reported instruments not named in the briefing — an emissions-trading expansion to major industrial sectors by 2027 and a National Low-Carbon

7. Structural Changes — Export Markets & the Supply Glut

Export markets are being reshaped by trade barriers. The US market has largely closed: tariffs as high as 3,521% were finalised on solar panels from Cambodia, Thailand, Malaysia and Vietnam — the routes Chinese firms used to reach the US — and Southeast Asia's share of US solar imports fell from 82% to 55% in 2024 (Caixin, Apr 2025). Europe remains open but is already China-supplied, sourcing about 97% of its solar equipment from China (Dialogue Earth, May 2025). The strategic response is to build factories abroad rather than just ship: Trina's chairman framed it as needing to localise production abroad because simply exporting is no longer enough (SCMP, Jun 2025), with new investment flowing to the Middle East (Saudi Arabia, Oman, Egypt), Indonesia and Laos (CSIS, Mar 2026).

On the supply glut, the main attempted fix has stalled. Six leading polysilicon producers proposed a roughly \$7bn fund to buy and idle about one-third of national polysilicon capacity, but China's market regulator (SAMR) halted the plan in early January 2026 over competition concerns and has not signalled a restart (pv magazine, Jan 2026). The softer lever is the government's 'anti-involution' campaign — pressure to stop selling below cost — launched July 2025, which (inferred, given the fund's suspension) is now the primary near-term mechanism. A further price factor: the 9% VAT export rebate is being removed from 1 April 2026 (Couleenergy, Jan 2026).

8. Medium-Term Outlook (5–10 Years)

The system will keep growing, but the manufacturing shake-out is the swing factor. Demand is durable and policy supports it: the plan calls for a 'new type of power system' able to absorb a high share of renewables, reinforced by an NDRC guideline targeting this by 2035 [15th FYP transcript, p.6; gov.cn, Feb 2026]. A concrete, primary-sourced build-out target is to push west-to-east power transmission capacity beyond 420 GW (420 million kilowatts), alongside major projects including a hydropower project on the lower Yarlung Zangbo River and new-energy bases in desert and Gobi areas [15th FYP transcript, p.5, p.17]. Non-fossil energy is targeted at 25% by 2030 (transcript p.17) and, per external analysis, 30% by 2035 (CREA, Mar 2026).

The unresolved questions are whether overcapacity is rationalised — which would restore manufacturing margins — and whether grid and storage build-out keeps pace with generation to limit curtailment. On storage, the target is large: 180 GW of new-energy storage by 2027, up from just over 100 GW at end-June 2025 (Carbon Brief, Oct 2025). A full demand/supply balance forecast is not directly stated in these sources and would be an inference, so it is flagged rather than asserted.

Investor Synthesis

Core economic engine. Two engines, not one — low-cost, high-scale manufacturing (currently margin-destroyed by oversupply) and power generation / operation (still profitable and revenue-stable). Value capture is migrating toward whoever owns the asset and the offtake, not the panel.

Primary growth lever. Sustained domestic electricity demand and electrification, backed by state planning — supplemented by exports increasingly routed through overseas factories rather than direct shipping.

Structural constraints investors underestimate. Curtailment and grid flexibility (capacity added is not power sold, especially in the northwest); the end of fixed tariffs under Document 136, which shifts price risk onto generators with patchy provincial rollout; and sticky overcapacity, since the flagship consolidation fund was blocked by the antitrust regulator, leaving margin recovery to slow voluntary attrition.

Key risk that could change trajectory. Trade barriers compounding — the US market is effectively closed and other regions are raising their own walls — combined with a failure to durably cut polysilicon and module oversupply, which would prolong manufacturer losses.

What kind of companies tend to win. *(Inferred from the financial and consolidation evidence above.)*

Lowest-cost producers with balance sheets to outlast the shake-out; vertically integrated players with profitable downstream generation; firms with genuine localised overseas manufacturing; and technology leaders positioned for the TOPCon-to-tandem transition. Sub-scale pure-play manufacturers without a cost or downstream edge are the structural losers.

Selected sources

- **Primary:** State Council Information Office briefing on the draft Outline of the 15th Five-Year Plan (2026–2030), NDRC, March 2026 — cited by page (p.5–p.21).
- National Energy Administration (NEA) 2025 capacity and output data, via gov.cn / Xinhua, Jan–Feb 2026.
- pv magazine; pv-tech; OPIS — polysilicon and module pricing, manufacturer H1/Q3 2025 results, consolidation plan and its suspension.
- InfoLink / TaiyangNews — 2025 module shipment rankings and export volumes.
- Centre for Research on Energy and Clean Air (CREA); Sightline Climate; Carbon Brief; Climate Action Tracker — 15th Five-Year Plan and climate targets.
- NDRC / NEA Document 136 (market-based pricing reform) coverage: pv magazine; pv-tech; Carbon Brief; InfoLink.
- CSIS; Caixin; SCMP; Dialogue Earth; Lowy Institute — export markets, US tariffs, overseas manufacturing.
- World Economic Forum — grid investment and flexibility (2024 data).
- CRU (via pv-tech); Interesting Engineering; SurgePV — cell-technology mix and tandem milestones.
- kr-asia — long-run electricity demand outlook.
- China Photovoltaic Industry Association (CPIA) 2024–2025 annual report figures, cited via CSIS.

Prepared as a factual industry overview to support independent analysis. Inferences are labelled in text; data vintage is noted where it is not current.